Divine Nwadigo

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Education

Core ML & MLOps — A Thinker's Comprehensive Program — Remote
Live cohort under Ayush Singh (Co-founder, SecondBrainLabs) — *May 2025 – 2026*

- Completed a live-lecture, cohort-based program covering the ML lifecycle end-to-end: data pipelines, model training, evaluation, and production deployment
- Practiced translating technical/ML work into clear communication for technical and non-technical audiences as part of program curriculum

Command Day Secondary School — Ibadan, Nigeria
Secondary School Certificate — *Aug. 2016 – June 2022*

Projects

Personalized Marketplace Discovery | *Python, PyTorch, FAISS, LightGBM, Airflow* — July 2026 – Aug. 2026

- Designed and built a two-stage recommendation system (retrieval + ranking) end-to-end for a simulated 2M-item marketplace catalog – from synthetic data generation and Spark/Airflow pipelines through model training, local serving, and a model promotion gate
- Built an item-anchored retrieval stage (sentence-embedding encoder + FAISS HNSW index, P99 <30ms) feeding a LightGBM LambdaMART ranking model that reorders candidates by purchase likelihood, within a 100ms end-to-end latency budget
- Evaluated retrieval and ranking independently against floor thresholds tied to downstream impact (Recall@200: 0.99 vs. 0.80 floor; NDCG@20: 0.39 vs. 0.25 floor) rather than optimizing offline accuracy in isolation
- Implemented a local champion/challenger promotion gate (MLflow Model Registry) that freezes a held-out evaluation set and requires a candidate model to clear both an absolute floor and the current production model before promotion – mirroring the manual-review step before a real A/B rollout
- Authored design docs (HLD/LLD) and scoped system constraints (catalog size, QPS, latency budget) before writing implementation code, documenting trade-offs made under those constraints (e.g., HNSW vs. IVF, single-node FAISS vs. distributed vector DB)

KKBox Churn Prediction System | *Python, SQL, DuckDB, Prefect, MLflow* — May 2026 – Aug. 2026

- Designed and built an end-to-end churn prediction system on a 970K-user subscription dataset (KKBox), decomposed into seven independently-invocable pipelines covering label derivation, feature engineering, model training/validation, daily batch scoring, and monitoring
- Built a two-track monitoring system – daily PSI-based feature-drift detection anchored to the training distribution, and monthly performance evaluation that automatically triggers a full retrain-and-validate pipeline when AUC-PR drops below threshold
- Translated model metrics into business terms by scoring churn risk 14 days before subscription expiry, a lead time sized to give CRM teams a real window to act before a subscriber actually churns
- Eliminated training-serving skew by routing training, serving, and monitoring through one shared feature module that enforces the point-in-time feature cutoff internally, closing the most common source of silent label leakage
- Documented deliberate infrastructure trade-offs made under solo-engineer constraints (Prefect over Airflow, DuckDB over Spark for feature aggregation, MLflow artifacts over DVC for dataset versioning), each backed by a written options-considered comparison

[**Project 3 Name**] | *[stack]* — [dates]

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Technical Skills

Languages: Python, SQL
ML / Modeling: PyTorch, LightGBM, sentence-transformers, FAISS, scikit-learn, pandas, NumPy
Data Engineering: PySpark, Apache Airflow, PostgreSQL, Snowflake
MLOps / Serving: MLflow (Model Registry), FastAPI, Docker, uv
Tools: Git